

VTPEH6270-CP04

Andrew Rodman

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1. Introduction

1.1 Packages

```
library(dplyr)
library(ggplot2)
```

1.2 Data Cleaning, Merging and Deriving

```
setwd("C:/Users/andre/OneDrive/Documents/VTPEH6270/")
fludata = read.csv("Data/Influenza_Laboratory-Confirmed_Cases_by_County__Beginning_2009-10_Season_20260")
svi = read.csv("Data/SVI_2016_US_county.csv")

flu_svi = merge(fludata, svi, by= "FIPS", all.x = T)
flu_svi$Count <- as.numeric(flu_svi$Count)
flu_svi$Week.Ending.Date <- as.Date(flu_svi$Week.Ending.Date, format = "%m/%d/%Y")
flu_svi$Season_Year <- as.numeric(substr(flu_svi$Season, 6, 9))
flu_svi = flu_svi[, c("Season_Year",
                      "Region",
                      "County",
                      "Week.Ending.Date",
                      "Disease",
                      "Count",
                      "E_TOTPOP",
                      "EP_POV",
                      "E_PCI",
                      "EP_UNEMP",
                      "FIPS")]
flu_svi <- flu_svi %>%
  rename(Total_Pop = E_TOTPOP,
         Poverty_Rate = EP_POV,
         Income_Per_Capita = E_PCI,
         Date = Week.Ending.Date,
         Unemployment_Rate = EP_UNEMP)

flu_svi_year <- flu_svi %>%
  group_by(Season_Year, County) %>%
```

```

summarise(
  Region = first(Region),
  Total_Count = sum(Count),
  Total_Pop = first(Total_Pop),
  Poverty_Rate = first(Poverty_Rate),
  Income_Per_Capita = first(Income_Per_Capita),
  Unemployment_Rate = first(Unemployment_Rate),
  FIPS = first(FIPS),
  .groups = "drop"
)
flu_svi_year <- flu_svi_year %>%
mutate(Incidence_Rate = (Total_Count / Total_Pop) * 100000)

```

2. Objective and Approach

2.1 Specific Objective

Do annual influenza incidence rates differ in New York State counties by income?

2.2 Data Description

```

variable_table <- data.frame(
  Name = c("County", "Income_Per_Capita", "Incidence_Rate", "Season_Year"),
  Type = c("Categorical", "Continuous", "Continuous", "Discrete"),
  Class = c("Character", "Numeric", "Numeric", "Date")
)

knitr::kable(variable_table, caption = "Variable Types and Classes")

```

Table 1: Variable Types and Classes

Name	Type	Class
County	Categorical	Character
Income_Per_Capita	Continuous	Numeric
Incidence_Rate	Continuous	Numeric
Season_Year	Discrete	Date

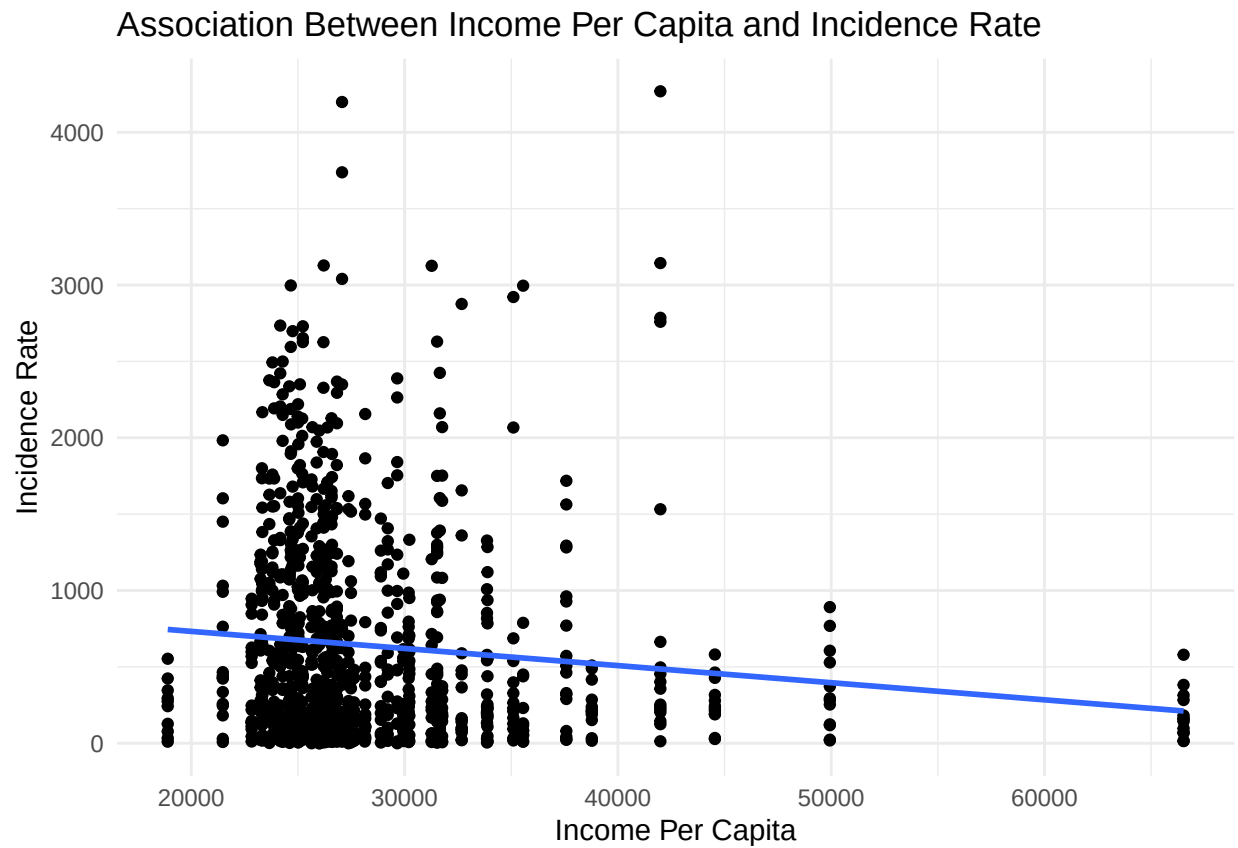
2.3 Data Visualization

```

ggplot(flu_svi_year, aes(x = Income_Per_Capita, y = Incidence_Rate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Association Between Income Per Capita and Incidence Rate",
    x = "Income Per Capita",
    y = "Incidence Rate"
  )

```

```
) +  
theme_minimal()
```



2.4 Description of Plausible Relationships

Incidence Rate and Income Per Capita seem to have a slightly negative linear relationship.

2.5 Parameters of Interest

```
parameters <- data.frame(  
  Parameter = c("Intercept", "Slope (Effect Size)", "Error / Noise"),  
  Symbol = c("a", "b", "epsilon"),  
  Description = c(  
    "Expected Incidence Rate when Income Per Capita = 0",  
    "Change in Incidence Rate for a one-unit increase in Income Per Capita",  
    "Random variation in Incidence Rate not explained by Income Per Capita")  
)  
  
knitr::kable(parameters, caption = "Parameters")
```

Table 2: Parameters

Parameter	Symbol	Description
Intercept	a	Expected Incidence Rate when Income Per Capita = 0
Slope (Effect Size)	b	Change in Incidence Rate for a one-unit increase in Income Per Capita
Error / Noise	epsilon	Random variation in Incidence Rate not explained by Income Per Capita

```
model <- lm(Incidence_Rate ~ Income_Per_Capita, data = flu_svi_year)
mean_income <- mean(flu_svi_year$Income_Per_Capita, na.rm = TRUE)
sd_income <- sd(flu_svi_year$Income_Per_Capita, na.rm = TRUE)
```

a = 956 b = -0.011 mean_income = 28987.5 sd_income = 7366

3. Simulation

3.1 Simulation Basis

```
set.seed(123)
n <- 200
Income_Sim <- rnorm(n, mean = 28987.5, sd = 7366)
a <- 956
b <- -0.011
epsilon <- rnorm(n, mean = 0, sd = 50)
Incidence_Sim <- a + b * Income_Sim + epsilon
sim_data <- data.frame(
  Income_Sim = Income_Sim,
  Incidence_Sim = Incidence_Sim
)
```

3.2 Simulation Output

```
sim_data$Income_Quartile <- cut(
  sim_data$Income_Sim,
  breaks = quantile(sim_data$Income_Sim, probs = seq(0, 1, 0.25)),
  include.lowest = TRUE,
  labels = c("Q1", "Q2", "Q3", "Q4")
)
summary_stats <- sim_data %>%
  group_by(Income_Quartile) %>%
  summarise(
    n = n(),
    Mean = mean(Incidence_Sim),
    Median = median(Incidence_Sim),
    SD = sd(Incidence_Sim),
    Min = min(Incidence_Sim),
    Max = max(Incidence_Sim)
  )
```

```
knitr::kable(summary_stats, caption = "Quartile Summary Statistics")
```

Table 3: Quartile Summary Statistics

Income_Quartile	n	Mean	Median	SD	Min	Max
Q1	50	741.2017	750.9837	60.49715	600.0246	872.3666
Q2	50	670.0379	667.3296	52.44653	555.5632	796.0612
Q3	50	612.2263	611.4604	40.02149	541.1322	700.3475
Q4	50	536.2860	542.4097	59.14110	404.5305	664.5536

3.3 Simulation Function

```
run_simulation <- function(
  effect_size = -0.011,
  noise       = 1.0,
  n           = 1000,
  a           = 956,
  mean_income = 28987.5,
  sd_income   = 7366,
  seed        = 42
) {
  set.seed(seed)
  sim_data <- data.frame(
    Income_Sim = rnorm(n, mean = mean_income, sd = sd_income)
  )
  sim_data$Incidence_Sim <- a + effect_size * sim_data$Income_Sim + rnorm(n, mean = 0, sd = noise)
  sim_data$Income_Quartile <- cut(
    sim_data$Income_Sim,
    breaks = quantile(sim_data$Income_Sim, probs = seq(0, 1, 0.25)),
    include.lowest = TRUE,
    labels = c("Q1", "Q2", "Q3", "Q4")
  )
  summary_stats <- sim_data %>%
    group_by(Income_Quartile) %>%
    summarise(
      n = n(),
      Mean_Incidence = mean(Incidence_Sim),
      Median_Incidence = median(Incidence_Sim),
      SD_Incidence = sd(Incidence_Sim),
      Min_Incidence = min(Incidence_Sim),
      Max_Incidence = max(Incidence_Sim),
      .groups = "drop"
    )
  list(
    params = list(effect_size = effect_size,
                  noise       = noise,
                  n           = n),
    sim_data = sim_data,
    summary_stats = summary_stats,
    table = knitr::kable(summary_stats, caption = "Quartile Summary Statistics")
  )
}
```

```
)
}
```

3.4 Simulation Automation

```
effect_sizes <- c(-0.001, -0.003, -0.005, -0.008, -0.011,
                 -0.015, -0.020, -0.030, -0.050, -0.080)

sample_sizes <- c(50, 100, 250, 500, 750, 1000, 2500, 5000, 10000, 50000)

noise_levels <- c(10, 50, 150)

param_grid <- expand_grid(
  effect_size = effect_sizes,
  n           = sample_sizes,
  noise       = noise_levels
)

simulation_results <- vector("list", nrow(param_grid))

for (i in seq_len(nrow(param_grid))) {
  simulation_results[[i]] <- run_simulation(
    effect_size = param_grid$effect_size[i],
    noise       = param_grid$noise[i],
    n           = param_grid$n[i],
    seed        = i
  )
}

names(simulation_results) <- paste0(
  "es=", param_grid$effect_size,
  "_n=", param_grid$n,
  "_noise=", param_grid$noise
)

all_summaries <- bind_rows(
  lapply(simulation_results, function(x) {
    x$summary_stats %>%
      rename(n_quartile = n) %>%
      mutate(
        effect_size = x$params$effect_size,
        noise       = x$params$noise,
        n           = x$params$sample_size
      )
  })
)

cat("Total simulation runs:", nrow(param_grid), "\n")
```

```
## Total simulation runs: 300
```

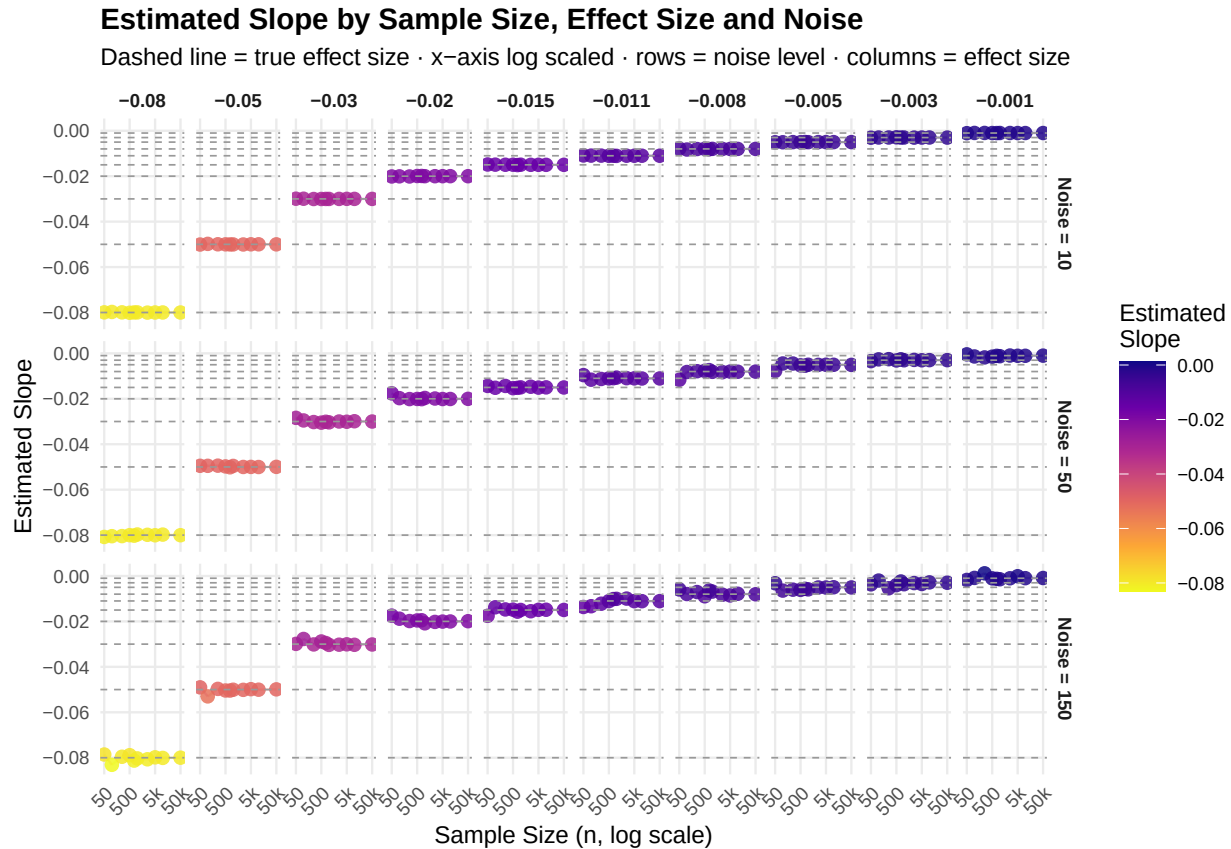
4. Visualization

```
slope_summary <- bind_rows(
  lapply(simulation_results, function(x) {
    if (is.null(x) || is.null(x$sim_data)) return(NULL)

    fit <- lm(Incidence_Sim ~ Income_Sim, data = x$sim_data)
    data.frame(
      effect_size = x$params$effect_size,
      sample_size = x$params$n,
      noise       = paste0("Noise = ", x$params$noise),
      est_slope   = coef(fit)[["Income_Sim"]]
    )
  })
)

slope_summary$noise <- factor(slope_summary$noise,
  levels = c("Noise = 10", "Noise = 50", "Noise = 150"))

ggplot(slope_summary,
  aes(x = sample_size, y = est_slope, colour = est_slope)) +
  geom_point(size = 2, alpha = 0.85) +
  geom_line(alpha = 0.4) +
  geom_hline(yintercept = as.numeric(as.character(
    unique(slope_summary$effect_size))),
    linetype = "dashed", colour = "grey60", linewidth = 0.3) +
  facet_grid(noise ~ effect_size) +
  scale_x_log10(
    breaks = c(50, 500, 5000, 50000),
    labels = c("50", "500", "5k", "50k")
  ) +
  scale_colour_viridis_c(option = "plasma", direction = -1,
    name = "Estimated\nSlope") +
  labs(
    title     = "Estimated Slope by Sample Size, Effect Size and Noise",
    subtitle  = "Dashed line = true effect size · x-axis log scaled · rows = noise level · columns = eff
    x         = "Sample Size (n, log scale)",
    y         = "Estimated Slope"
  ) +
  theme_minimal(base_size = 9) +
  theme(
    strip.text      = element_text(face = "bold", size = 7),
    plot.title      = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    axis.text.x     = element_text(angle = 45, hjust = 1, size = 7),
    legend.position = "right"
  )
)
```



5. Interpretation

These simulations showcase that the effect is consistently recovered across parameters, meaning that the estimated slope is always negative, consistent with our original data. The direction never reverses. Noise also has little influence, as all 3 noise distributions, look close. The quartile gradient is also real and detectable, supporting the association between income, and influenza incidence.

6. AI Use Disclosure Statement

As part of this assignment, please indicate whether you used any AI-based tools (e.g., ChatGPT, Claude, Copilot, Gemini, etc.). Indicate: • Did you use AI? Yes

- If yes: Write a short disclosure statement (2–3 sentences) describing: • Which tool(s) you used. Claude and ChatGPT
- How you used it (e.g., to help decide on an analysis approach, to generate a first draft of code, to improve or debug code you had written yourself, etc.). Example: “This document was generated using Claude to generate original code, which was then reviewed and adjusted” or “This document was generate using ChatGPT to assist with debugging of code generated by the author”. Please keep your statement brief and honest. The goal is transparency, not detail: we do not need exact prompts or transcripts, just a clear sense of how AI supported this work

This document was generated using Claude and ChatGPT to formulate a simulation, and produce graphs and the multi-panel grid.